HAND & UPPER LIMB · NEET-SS / FET / INI-SS

nerve compression and injury

Nerve injury

Seddon	Sunderland	Pathology	Recovery
Neurapraxia	I	Conduction block, axon intact	Complete, weeks
Axonotmesis	II-IV	Axon (and progressively endoneurium, perineurium) disrupted, epineurium intact	II-III good; IV needs surgery
Neurotmesis	V	Complete transection	Needs repair

- Mackinnon added a mixed **type VI**.
- **Wallerian degeneration** distal to the injury. Regeneration is about **1 mm/day (1 inch a month)**; an **advancing Tinel sign** tracks it.
- Motor end plates deteriorate after about **12-18 months**, so motor recovery needs reinnervation before then. This sets the timing of repair, grafts and transfers.
- **Repair:** tension-free epineural (or group fascicular) repair. Use a **sural nerve graft** for gaps; **conduits** only for short sensory gaps (under about 3 cm).

Median nerve

- **Carpal tunnel syndrome** (commonest compression neuropathy):
  - Night pain and paraesthesia in the radial 3.5 digits. Durkan compression, Phalen and Tinel tests.

  - **Thenar sensation is spared**: the palmar cutaneous branch runs superficial to the transverse carpal ligament. Sensory loss over the thenar eminence points to a more **proximal** lesion.
  - Nerve conduction: prolonged **distal motor latency** (over about 4.2-4.5 ms) and slowed sensory conduction.
  - Treat with a neutral night splint and steroid injection, then **open or endoscopic release**. The recurrent motor branch is usually extraligamentous.
- **Pronator syndrome:** aching in the proximal forearm, sensory loss including the palm. Compression sites: ligament of Struthers, **lacertus fibrosus**, between the two heads of **pronator teres**, FDS arch. Nerve studies are often normal.
- **AIN syndrome: pure motor** (FPL, FDP to index and middle, pronator quadratus). The patient **cannot make an "OK" sign** (pinch becomes pulp-to-pulp).

Ulnar nerve

- **Cubital tunnel** (second commonest):
  - Compression sites: arcade of Struthers, medial intermuscular septum, medial epicondyle, **Osborne ligament** (commonest), between the FCU heads, deep flexor-pronator aponeurosis.
  - Signs: **Froment** (FPL substitutes for adductor pollicis), **Wartenberg sign** (abducted little finger), Jeanne sign, intrinsic wasting.
  - The **dorsal ulnar cutaneous branch is involved**, which distinguishes it from Guyon's canal.
  - **Ulnar paradox:** clawing is worse with a distal lesion (FDP to the ring and little fingers is spared).

  - **In situ decompression** gives results similar to anterior transposition. Transpose if the nerve subluxes.
- **Guyon's canal zones:** I mixed motor and sensory (proximal to the bifurcation); II motor only (deep branch; **hook of hamate** fracture, ganglion); III sensory only. A **ganglion** is the commonest cause. Hypothenar hammer syndrome: ulnar artery thrombosis.

Radial nerve

- **PIN syndrome: motor only**, finger and thumb drop. The wrist extends with **radial deviation**, because ECRL is spared and ECU is weak.
  - Compression sites: fibrous bands at the radiocapitellar joint, **leash of Henry**, ECRB edge, **arcade of Frohse** (commonest), distal supinator edge.
  - **Radial tunnel syndrome** is pain at the same sites without weakness.
- **Wartenberg syndrome:** superficial radial nerve between brachioradialis and ECRL, worse with pronation.
- **Radial palsy with a humeral shaft fracture** (classically the distal-third spiral **Holstein-Lewis** fracture): mostly neurapraxia. **Observe 3-4 months**; explore early for open fractures, and later if there is no recovery.
- **Saturday night palsy** (spiral groove): wrist drop with **triceps spared**.

Written by Claude for Ortho preSSaleel · updated 2026-11-14. Check doubtful points in a standard textbook.